

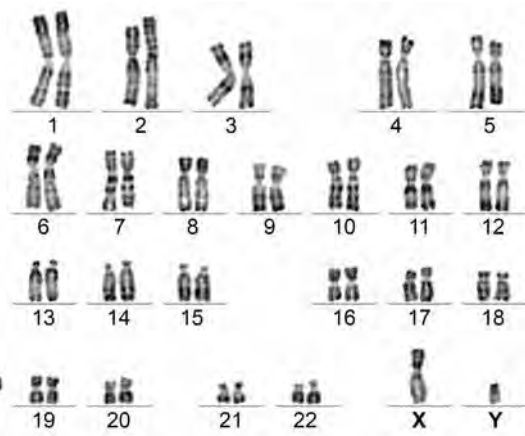
0 1

Figure 1 shows all the chromosomes present in one human cell during mitosis. A scientist stained and photographed the chromosomes. In **Figure 2**, the scientist has arranged the images of these chromosomes in homologous pairs.

Figure 1



Figure 2



0 1 . 1

Give **two** pieces of evidence from **Figure 1** that this cell was undergoing mitosis. Explain your answers.

[2 marks]

1. _____

2. _____

0 1 . 2

Tick (✓) **one** box that gives the name of the stage of mitosis shown in **Figure 1**.

[1 mark]

Anaphase

☐

Interphase

☐

Prophase

☐

Telophase

☐

0 1 . 3

When preparing the cells for observation the scientist placed them in a solution that had a slightly higher (less negative) water potential than the cytoplasm. This did not cause the cells to burst but moved the chromosomes further apart in order to reduce the overlapping of the chromosomes when observed with an optical microscope.

Suggest how this procedure moved the chromosomes apart.

[2 marks]

Question 1 continues on the next page

Turn over ►

0 1 . 4

The dark stain used on the chromosomes binds more to some areas of the chromosomes than others, giving the chromosomes a striped appearance.

Suggest **one** way the structure of the chromosome could differ along its length to result in the stain binding more in some areas.

[1 mark]

0 1 . 5

In **Figure 2** the chromosomes are arranged in homologous pairs. What is a homologous pair of chromosomes?

[1 mark]

0 1 . 6

Give **two** ways in which the arrangement of prokaryotic DNA is different from the arrangement of the human DNA in **Figure 1**.

[2 marks]

1.

2.
